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Project Title:	Multisensory Interactions to Discriminate Gravity From Translational Acceleration		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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Human Research Program Risks:	None		
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Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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	Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will characterize how gravity and translational accelerations are encoded by vestibular neurons in the cerebellum of alert primates. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely		

Task Description:	incomplete. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques. Aim I: Investigate the vestibular convergence patterns in the rFN and NVU: We propose to use combinations of three-dimensional (3D) rotations and translations in order to characterize the otolith/semicircular canal convergence patterns in the rFN and NVU. Specifically we propose: (1) To examine if there are populations of rFN and NVU neurons with properties similar to those found in the VN (Canal-Only, Otolith+Canal and Otolith-Only cells). (2) To quantify differences and/or similarities in the 3D distributions of rotational maximum sensitivity vectors using earth-vertical axis (EVA) rotation (canal only stimulation), as well as in the spatial distributions of translational sensitivity vectors (otolith only stimulation) between convergent and non-convergent cells. (3) To compare EVA rotation and translational responses with cell firing rate modulation during earth-horizontal axis (EHA) rotations in order to examine the pattern of otolith/canal convergence in the rFN and NVU. Our working hypothesis is that the rFN and NVU circuitry is specific to the processing of linear acceleration. Thus, unlike the VN where a quarter of the sampled neuronal population was identified as Canal-Only cells, we hypothesize that we will encounter very few, if any, Canal-Only neurons in the rFN and NVU. Aim II: Test if rFN and NVU neurons discriminate dynamic tilts relative to gravity from translational components of linear acceleration: Although primary otolith afferents respond equally to translational acceleration and dynamic tilts relative to gravity, central vestibular neurons could selectively encode the translational component of the motion. We will quantitatively test this hypothesis using protocols with combinations of translational and tilt motion stimuli. We hypothesize that a subpopulation of rFN and NVU neurons, including all neurons antidromically identified as projecting to the VN, will selectively respond to translational head motion and ignore changes in head orientation relative to gravity. Aim III: Investigate the coordinate system (head vs. body-fixed) for encoding translational motion by rFN and NVU neurons: Primary otolith afferents encode linear acceleration in a head-fixed coordinate system. However, spatial orientation, postural control and self-motion awareness might require that there exist central neurons encoding translational motion of the body in space. In these studies we will investigate the influence of static neck inputs in the processing the otolith information by systematically varying both the direction of sled motion relative to the body and the static orientation of the head relative to the trunk. Because both coordinate systems might be important for downstream processing, we hypothesize that translation-coding rFN and NVU neurons that project to the VN include both head-, as well as body-fixed coordinate systems. Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will characterize how gravity and translational accelerations are encoded by vestibular neurons in the cerebellum of alert primates. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely incomplete. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed studies are also directly relevant to understanding complex processing and multisensory interactions in the brain. In addition to space flight, this knowledge is important for understanding brain function and computational neural networks.
Task Progress:	As creatures living in a gravitational environment, the force of gravity represents an important variable for both spatial perception and sensorimotor transformations. In particular, the direction of gravity provides a veridical allocentric frame of reference, important for identifying our own motion relative to the inertial (i.e., earth-centered) world. However, inertial motion information is not directly available in vestibular afferent activities, because of two sensory ambiguities. The first, referred to as the ‘rotation problem’, arises because of the physical properties of the semicircular canals: Primary canal afferents detect endolymph fluid motion relative to the bony ducts, coding rotation in a head-centered reference frame and providing no information about how the head moves relative to the outside world. For example, a horizontal (yaw) rotation in upright orientation activates semicircular canal afferents similarly as a yaw rotation in supine orientation. Yet, these two movements differ in allocentric (earth-centered) space. The second ambiguity, referred to as the ‘linear acceleration problem’, arises because of the physical properties of the otolith organs, which detect net linear acceleration and cannot distinguish translational from gravitational components. The brain thus faces the daunting task of estimating inertial motion and spatial orientation, an important component of which is the segregation of gravitational and translational accelerations. Other than when perceptual illusions occur (“oculogravic illusion”), these salient reference frame transformations do not often reach perceptual awareness because under most everyday conditions there exists a unique mathematical solution. First, rotational signals from the semicircular canals (coded in head-centered coordinates) must be processed by a head orientation relative to gravity signal to construct an estimate of angular velocity in earth-centered coordinates. Such a transformation decomposes angular velocity into two spatially-defined components: an earth-vertical (i.e., parallel to gravity) component and an earth-horizontal (perpendicular to gravity) component. Importantly, only the latter component signals an orientation change relative to gravity, i.e., the change in spatial altitude (‘tilt’) relative to the outside world. In a second computational step, net linear acceleration from the otolith organs can be combined with this ‘tilt’ signal to extract the linear acceleration component that is due to translation. What is needed is an angular orientation signal, which can be computed by temporal integration. Notice that these two computational problems are not independent, but intimately coupled, as rotational velocity cannot be interpreted in a spatial frame without knowledge of head orientation relative to gravity. Similarly, parsing net linear acceleration into orientation (gravitational) and translational acceleration components requires knowledge about the integral of rotational velocity in an earth-centered reference frame. Because perceptual illusions rarely occur in everyday activities, even in darkness, inertial motion can be likely computed using inner ear information alone. A population solution might utilize a brainstem/cerebellar network that includes vestibular and medial cerebellar nuclei neurons. Both of these areas receive strong bi-directional projections from the posterior vermal lobules (lobule X: nodulus; lobule IX: ventral uvula, collectively referred to as NU), which also receive direct primary vestibular afferent projections. During this grant year we have tested the hypothesis that the inertial motion computations occur in the

	cerebellar cortex by quantifying the simple spike activities of NU Purkinje cells. We found that, unlike vestibular and cerebellar nuclei neurons, NU Purkinje cells represent a homogenius population that encodes motion in an inertial reference frame. These results constrain the processing of gravitoinertial accelerations and supports the hypothesis that the necessary computations for inertial motion and orientation occur within the cerebellar cortex.
Bibliography Type:	Description: (Last Updated: 04/21/2017)
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Articles in Peer-reviewed Journals	Shaikh AG, Green AM, Ghasia FF, Newlands SD, Dickman JD, Angelaki DE. "Sensory convergence solves a motion ambiguity problem." Curr Biol. 2005 Sep 20;15(18):1657-62. PMID: 16169488 , Sep-2005